

Problem 25.14

A company makes payments continuously at a rate of \$200 per year. The payments are made between 2 and 7 years. Find the accumulated value of these payments at time 10 years using an annual rate of interest of 6.5%.

Solution

The payments are made continuously from time 2 to time 7, so the payment period has length

$$7 - 2 = 5$$

years.

The annual effective interest rate is

$$i = 0.065.$$

The corresponding force of interest is

$$\delta = \ln(1 + i) = \ln(1.065).$$

The accumulated value at time 7 of the continuous payments is

$$200\bar{s}_{\overline{5}|}.$$

Then this value is accumulated from time 7 to time 10, which is 3 more years. Thus the accumulated value at time 10 is

$$AV_{10} = 200\bar{s}_{\overline{5}|}(1 + i)^3.$$

For a continuous annuity,

$$\bar{s}_{\overline{5}|} = \frac{e^{5\delta} - 1}{\delta}.$$

Since $e^\delta = 1.065$,

$$e^{5\delta} = (1.065)^5.$$

Therefore,

$$AV_{10} = 200 \left(\frac{(1.065)^5 - 1}{\ln(1.065)} \right) (1.065)^3.$$

Thus,

$$AV_{10} = 1419.761721.$$

So the accumulated value at time 10 is

$$\boxed{1419.76}.$$